AI Patent Production by Region

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***Abstract*— This study examines how major global regions convert AI research and investment inputs into patent output using a longitudinal panel from 2013–2023. Data from the Stanford AI Index and World Bank indicators are used to analyze AI publications, private investment, and economic scale across four regions. Exploratory analysis shows rapid growth in AI patent activity in Asia relative to the United States, Europe, and India. A region and year fixed-effects model finds that AI research publications are a statistically significant predictor of patent output, while private AI investment is not significant once fixed effects are included. These results suggest that sustained research capacity, rather than short-term investment fluctuations, is a primary driver of AI patent production across regions.**

I. Introduction

Artificial intelligence, or AI, is an important tool that's seeing a surge of investment and focus in both private and public spheres. In November of 2025, the White House announced the Genesis Mission as an executive order of the president. Genesis Mission is an initiative to coordinate efforts to advance, revolutionize, and develop AI. The mission is to harness AI to accelerate innovation and technological discovery. The United States is not alone in pursuing AI, as countries around the world are getting into the AI race, which has been referred to as "comparable in urgency… to the Manhattan Project" [4].

This research was motivated after a literature review identified differences in patent production between the US and China in machine vision. The chart below from a report indicates that most AI patents in China are coming from universities, whereas in the US the majority comes from industry [3]. China also surpasses the US in AI patent production, but the mechanisms through which they produce patents are different.

Regions such as India and Europe also produce a relatively low quantity of patents compared to Asia and the US. Understanding these disparities requires examining how countries convert inputs such as research output, R&D investment, education, and private sector funding into measurable innovation outcomes.

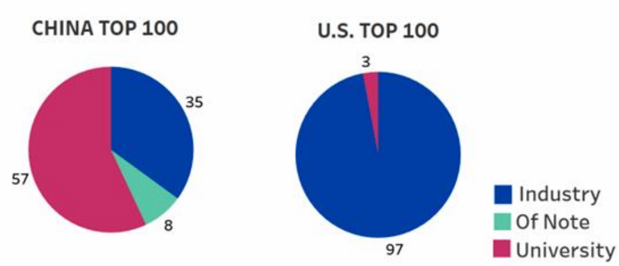

Figure 1: Private vs Public Patent Production (US/China) [3]

There are several fields that fall under the umbrella of AI, and this study broadly considers all of them including machine learning, computer vision, robotics, and generative AI. China's primary technological and patent focus has been on computer vision. Identifying which factors account for the difference in patent production across regions could benefit policymakers and technical leaders that want to drive technical innovation.

The following hypotheses will be tested in the Methods section:

H_{01}: After controlling for region and time, AI publications have no association with patent output.

H_{11}: AI publications are positively associated with patent output.

H_{02}: Private investment has no association with patent output once region fixed effects are included.

H_{12}: Private investment is positively associated with patent output.

The hypotheses are tested using a region and year-fixed effects model.

II. DATASETS

The primary datasets used in this report were the Stanford AI Index Report 2025, and the World Bank World Development Indicators (WDI) [1] [2]. The Stanford Report is the eighth edition of this report published by researchers at Stanford Institute for Human-Centered AI (HAI). It is an institute dedicated to advancing AI research, education, and policy.

The Stanford report has categories including R&D, technical performance, responsible AI, economy, science and medicine, policy, education, and public opinion. To limit the scope of the study, this report is primarily focused on metrics within the R&D and economy sections of the report. These factors were chosen because the motivation is to understand R&D patent output differences between different geographical regions based on factors such as national R&D, and private sector investment.

In the dataset, the data are primarily grouped by geographic region. Some categories did not have data available for all regions, especially those that are low producers of AI patents - Latin America and the Caribbean, Middle East and North Africa, and Sub-Saharan Africa. For this reason, the regions that were studied were the Asia, Europe, India, and the US.

Using this dataset, I investigated three questions: How do AI research output, R&D investment, and private investment relate to AI patent activity across major economies? Do countries follow similar or divergent trajectories in their AI innovation? To what extent do region-level factors, beyond measurable inputs, explain differences in AI innovation?

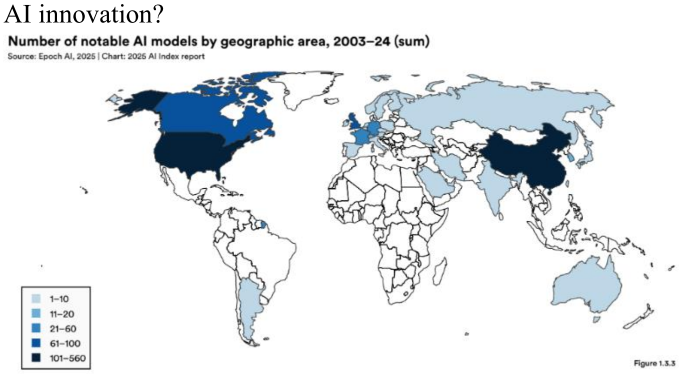

Figure 2: AI Models by Geographic Area

The final data panel has the following longitudinal data from 2013-2023:

Table 1: Data Panel

Factor	Data Type
Year	Integer
Publications (% of global total)	Numeric
Publications (global total)	Numeric
Total Publications (per region)	Numeric
Citations (% of global total)	Numeric
Patents (% of global total)	Numeric
Private Investment (USD billions)	Numeric
GDP (per region)	Numeric
R&D Expenditure (USD billions)	Numeric

Some data was available before 2013, but 2013 was selected as the earliest year for the repeated measure data panel because it is the first year where there is data available for all regions. There was a total of n = 11 years available. Some factors, such as R&D spending, were only available for two regions (Asia and US), limiting their usefulness as a predictive factor. The final panel consists of 4 regions observed over 11 years, for a total of N = 44 region-year observations.

III. METHODS

A review of the relevant sections of the report identified several factors of interest. The Stanford Report includes csv files of longitudinal data for the following:

Publications (global total)
Publications (% of global total per region)
Citations (% of global total)
Patents (% of global total)
Private investment (USD)

The World Bank data includes the following:

GDP (per region)
R&D expenditure (USD)

From the available data, the total number of publications per region was calculated:

Total publications (per region) = Publications (global total) * Publications (% of global total per region)

All the data is longitudinal data, grouped by region, and available from 2013-2023. AI patents granted per region was selected as a useful standard innovation indicator for evaluating national AI capability.

I started with some exploratory plots to begin to visualize and understand the data.

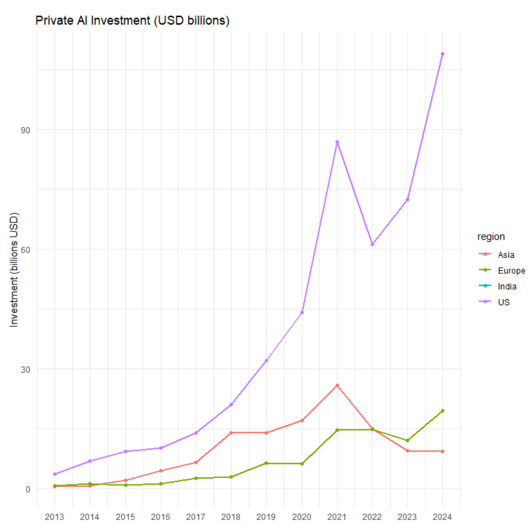

Figure 3: Private AI Investment (USD billions)

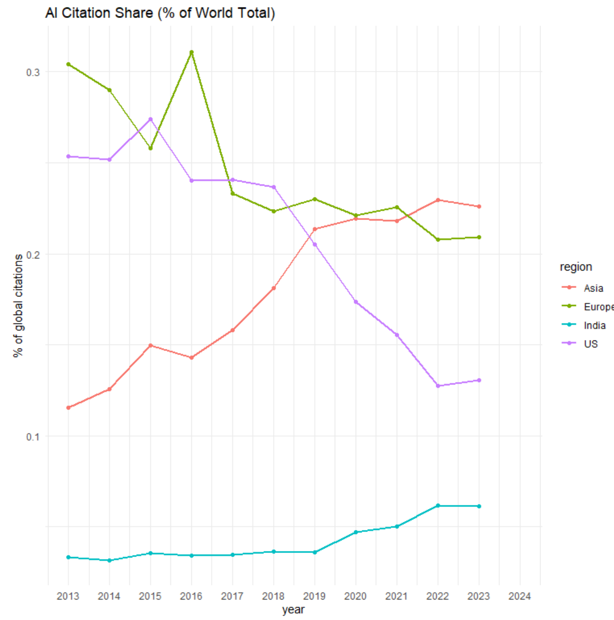

Figure 4: AI Citation Share (% of World Total)

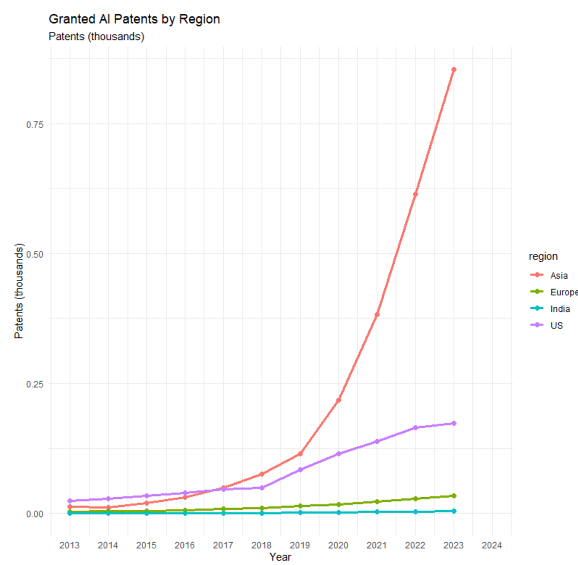

Figure 5: Granted AI Patents by Region

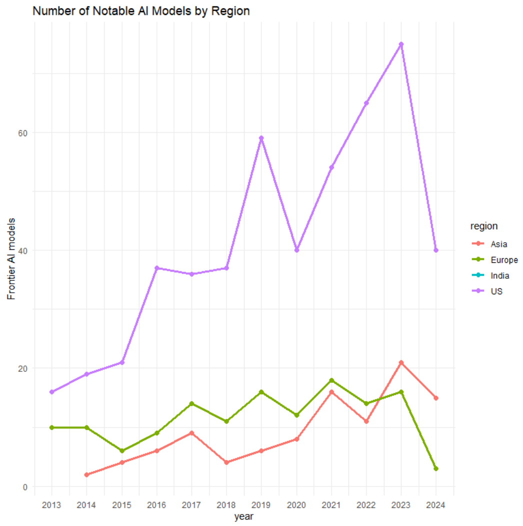

Figure 6: Number of Notable AI Models by Region

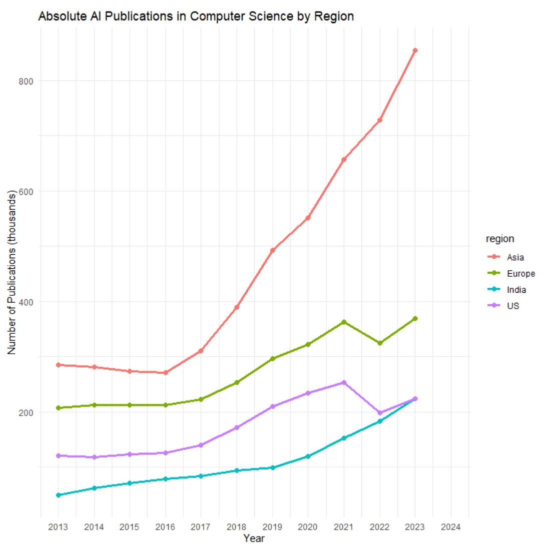

Figure 7: Absolute AI Publications by Region

It is notable that Asia (primarily driven by China) is seeing rapid exponential growth of patent output, whereas the US has seen a moderate increase in patent output along with Europe and India (Figure 5). Private investment over the same period is increasing, where the US leads by far and China is seeing a decrease in private investment in recent years (Figure 3). The US also leads in notable AI models (most of which are produced by private industry – Figure 6). The initial visualizations seem to indicate that there are distinct innovation models (public vs private) that are driving AI production in different regions. This is purely a descriptive analysis that will be formally tested later.

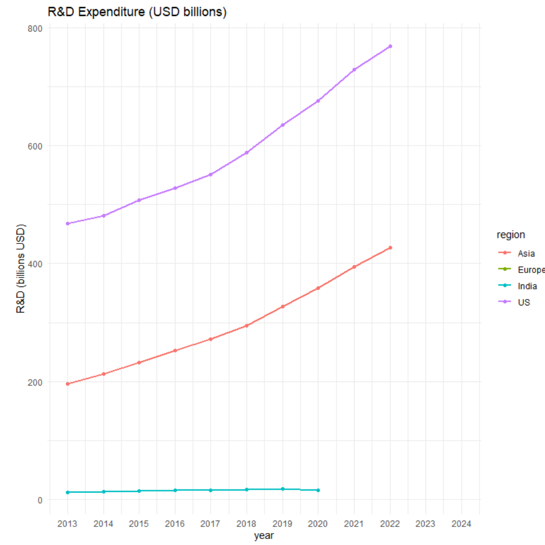

Figure 8: R&D Expenditure (USD billions)

All regions are seeing an increase in AI publications over time (Figure 7), though the overall share of citations is increasing for China and India, and decreasing for the US and Europe (Figure 4). As for R&D expenditure, available data only for China and the US indicates increasing expenditure over time (Figure 8). All visual trends are descriptive and helped with formal evaluation in the regression models below.

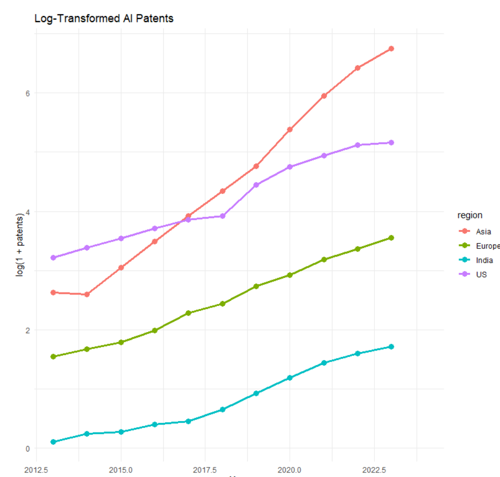

Figure 9: Log-transformed AI Patents

Due to the exponential growth of patent production in Asia, I decided to do a logarithmic transformation of all data to convert exponential growth into linear growth, stabilize the variance, and reduce skewness and heteroscedasticity of models. The data will also be interpretable as elasticities, percentage changes. Figure 9 shows the transformed data.

I began the analysis by treating all years and regions as independent and creating an ordinary least squares model using total AI patents as the dependent variable, and publications, GDP, and year as factors.

While the model did have an R^2 of 0.712, and log(GDP) was found to be significant at the $\alpha = .05$ level, the model violates several key linear model assumptions that make it invalid for this type of data (Figure 10):

Heteroscedasticity: there is a curved pattern in the residuals
Normality: the Q-Q plot shows both left skew and heavy tails
Independence: the data violates the assumption of independence because it is longitudinal

Next, I decided to investigate a fixed-effects model as a better fit.

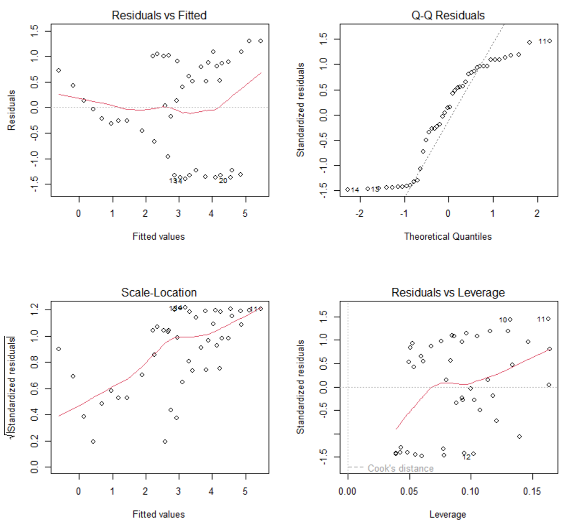

Figure 10: Ordinary Least Squares Diagnostics

To build a fixed-effects model, I started with the available factors of interest and added them one-by-one to identify the best model (Figure 11). The dependent variable was log(patents), and factors were log(publications), log(gdp), log(investment), and log(rd). The models were evaluated based on AIC and within R^2, where a lower AIC estimates a lower prediction error for the model. Though model 4 has the lowest AIC, after checking multicollinearity, GDP and R&D spending were found to be highly collinear, so R&D spending was dropped as a factor (Figure 12). Model 3 was the best model found with a within R^2 value of 0.94, meaning that 94% of the variation in patent output is explained by the model.

	m1	m2	m3	m4
Dependent Var.:	log_patents	log_patents	log_patents	log_patents
log_pubs	0.8078 (0.5192)	-0.6070 (0.6944)	0.9777* (0.4454)	1.933** (0.3932)
log_gdp		3.921* (1.421)	4.624*** (0.7056)	11.11* (2.984)
log_inv			0.0081 (0.0969)	0.2996* (0.1059)
log_rd				-9.189. (3.889)
Fixed-Effects:	---------------	---------------	---------------	---------------
region	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
S.E. type	IID	IID	IID	IID
AIC	58.994	50.404	-35.024	-44.704
BIC	85.757	78.951	-11.079	-29.768
R2	0.96189	0.97004	0.99560	0.99866
Within R2	0.07705	0.27448	0.93982	0.98732

Figure 11: Iterative Modeling

	log_rd	log_gdp	log_inv	log_pubs
log_rd	1.0000000	0.9887233	0.82473029	-0.40478303
log_gdp	0.9887233	1.0000000	0.83882467	-0.37833224
log_inv	0.8247303	0.8388247	1.00000000	0.07485766
log_pubs	-0.4047830	-0.3783322	0.07485766	1.00000000

Figure 12: Correlation Table

The fixed-effects model has the form:

$$\log(\text{patents})_{it} = \beta_1 \log(\text{pubs})_{it} + \beta_2 \log(\text{gdp})_{it} + \beta_3 \log(\text{investment})_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Table 2: Fixed Effects Model Coefficients

Factor	β	p
log(publications)	0.978	.042
log(gdp)	4.624	<.001
log(investment)	.008	0.93

The coefficient of log(publications) is positive and statistically significant, indicating a near one-to-one elasticity between research output and patent production. In contrast, private investment does not exhibit a statistically significant within-region relationship with patent output once fixed effects are included. GDP has a large coefficient, and a low p-value, but this large elasticity may indicate that there are omitted scale-related factors rather than causal effects. The within R^2 value of the model is 0.94, meaning a large portion of the variability in patent output is explained by the model.

Hypothesis Testing Results:

H_{01} rejected (p = 0.042)
H_{02} not rejected (p = 0.93)

The null hypothesis that AI publications have no association with patent output (H_{01}) is rejected, indicating a statistically significant positive relationship between research output and AI patent production. In contrast, the null hypothesis that private investment has no association with patent output (H_{02}) cannot be rejected. These results suggest that AI patent growth is more closely tied to research capacity than private investment.

IV. Discussion and Conclusion

This study examined the relationship between AI research output, economic scale, and private investment with AI patent production across major global regions using a longitudinal fixed-effects model. The results indicate that AI publications are a statistically significant and economically meaningful predictor of AI patent output within regions over time, while private AI investment is not statistically significant once region and year fixed effects are included.

The rejection of H_{01} suggests that increases in research activity are closely associated with corresponding increases in patent production. The near one-to-one elasticity between publications and patents implies that regions that expand their AI research output tend to translate that research into patentable innovations at a roughly proportional rate. This finding is consistent with the view that scientific capacity and knowledge creation are foundational drivers of downstream innovation, regardless of whether that research originates in universities or industry.

In contrast, the failure to reject H_{02} indicates that private AI investment does not explain within-region changes in patent output over time after controlling for unobserved regional characteristics and global time trends. This result does not imply that private investment is unimportant for AI innovation, but it suggests that differences in private investment across regions may be explained by underlying structural factors captured by the region fixed effects. In other words, private investment may help explain why some regions consistently produce more patents than others, but it does not appear to drive year-to-year changes in patent output within a given region.

The large and statistically significant coefficient on GDP further supports the interpretation that overall economic scale and institutional capacity play a substantial role in shaping innovation outcomes. However, GDP likely masks multiple omitted factors, including labor force size, education systems, and institutional support for innovation. Its estimated elasticity should not be interpreted as a purely causal effect.

Together, these findings suggest that AI innovation systems differ not only in magnitude but also in mechanism. Regions characterized by strong research ecosystems appear better able to convert knowledge production into patentable outcomes, while short-term fluctuations in private investment alone are insufficient to explain changes in patent output. This distinction is particularly relevant when comparing regions with predominantly public-sector-driven research models to those led by private industry.

Limitations

Several limitations should be considered when interpreting these results. First, the analysis is conducted at the regional level, which masks substantial heterogeneity at the country, firm, and institutional levels. Aggregating data across diverse economies such as those within Asia or Europe may obscure important national differences in innovation systems.

Second, AI patents are an imperfect proxy for innovation quality or economic impact. Patent counts do not capture differences in patent scope, influence, or downstream commercialization, nor do they reflect non-patented innovations such as open-source models or trade secrets.

Third, data availability constrained the inclusion of certain variables, particularly R&D expenditure, which was only available for a subset of regions. Although multicollinearity concerns justified its exclusion from the final model, the absence of comprehensive R&D data limits the ability to fully disentangle the roles of public and private investment.

Conclusion

This study provides evidence that AI patent production across major global regions is more strongly associated with sustained research output than with short-term changes in private investment once regional and temporal factors are accounted for. The results highlight the central role of scientific capacity in driving innovation and suggest that policies aimed at expanding and maintaining research ecosystems may be more effective at increasing patent output than policies focused solely on stimulating private investment.

These findings have implications for policymakers and research institutions seeking to enhance national AI competitiveness. Investments in education, research infrastructure, and knowledge creation appear to be critical components of long-term innovation capacity. Future research could extend this analysis by incorporating country-level or firm-level data, alternative innovation metrics such as citation-weighted patents or model performance measures, and more granular measures of public and private R&D investment.

References

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